# Glivenko's theorem underneath structure

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## Abstract

Glivenko's theorem has been studied in an abstract setting, where double negation is replaced by arbitrary nuclei, and classical and intuitionistic propositional logics are generalised to abstract entailment relations. This paper extends that framework to substructural logics, focusing on cases where structural rules such as weakening, contraction, and exchange are restricted. We show that while the absence of weakening or contraction has minimal impact, the absence of the exchange rule introduces significant and nontrivial distinctions.

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## 1 Introduction

Glivenko's theorem states that in propositional logic, the classical provability of a formula implies the intuitionistic provability of its double negation [11, 12]:

$$\mathbf{Cla} \vdash U \rhd a \iff \mathbf{Int} \vdash U \rhd \neg\neg a,$$

where **Cla** and **Int** denote classical and intuitionistic propositional logic, respectively.

Various generalisations of this theorem have been developed by replacing **Int** with weaker logics. In 1968, Segerberg [32] introduced the *Glivenko logic* **Gli**, defined as Johansson's minimal logic extended by the double negation of *ex falso quodlibet sequitur*, namely $\neg\neg(\bot \to a)$. In the same paper, he claimed—without proof or citation—a characterisation result that was later formally established by Odintsov [24, 25]: the Glivenko logic is the smallest logic for which Glivenko's theorem holds. Specifically, for any logic $\mathbf{L}$ such that $\mathbf{Min} \subseteq \mathbf{L} \subseteq \mathbf{Cla}$, the following are equivalent:

1. $\mathbf{Cla} \vdash U \rhd a$ if and only if $\mathbf{L} \vdash U \rhd \neg\neg a$;
2. $\mathbf{L}$ is an extension of **Gli**.

Extensions of this result to substructural logics were developed by Cignoli and Torrens [4, 5], and by Galatos and Ono [10, 9]. Later, Ono [27] proved a version of Segerberg's characterisation for the Full Lambek logic with exchange, $\mathbf{FL}_e$: for any $\mathbf{L}$ such that $\mathbf{FL}_e \subseteq \mathbf{L} \subseteq \mathbf{InFL}_e$, the following are equivalent:

1. $\mathbf{InFL}_e \vdash U \rhd a$ if and only if $\mathbf{L} \vdash U \rhd \neg\neg a$;
2. The following conditions are satisfied:
   - $\mathbf{L} \vdash \neg(a \wedge b) \rhd \neg(\neg\neg a \wedge \neg\neg b)$;
   - $\mathbf{L} \vdash \neg(a \to b) \rhd \neg(\neg\neg a \to \neg\neg b)$.

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We refer to any logic satisfying these conditions as the *Glivenko extension* of $\mathbf{FL}_e$.

Double negation in intuitionistic logic is a specific instance of a *nucleus* [2, 14, 15, 22, 30, 33, 34, 3].—also referred to as a strong monad [17] or lax modality [1]—defined as a map $j$ on formulae satisfying:

$$(a \to j(b)) \lhd\!\rhd (j(a) \to j(b)).$$

Fellin and Schuster [7, 8] generalised Glivenko's theorem by replacing double negation with an arbitrary nucleus, transitioning from syntactic provability in a calculus to an inductively generated abstract consequence relation, and extending the framework from propositional formulas to arbitrary objects. They provided precise criteria for when a classical generalisation becomes a conservative extension of its intuitionistic counterpart under double negation.

This paper further generalises that framework to entailment relations that lack structural rules such as weakening, contraction, and exchange [28, 29]. We show that the absence of weakening or contraction has limited effect on the overall picture, while the absence of exchange introduces significant and nontrivial distinctions. One key insight is that it is not necessary for $j$ to be a nucleus over the starting logic $\mathbf{L}$: it suffices for $j$ to be merely expansive over $\mathbf{L}$ and to become a nucleus in its Glivenko extension. In traditional settings, the presence of exchange and implication ensures that $j$ remains a nucleus over all inductive extensions of $\mathbf{L}$, including the Glivenko extension. However, in the general substructural setting, this no longer holds, making the initial nucleus assumption dispensable.

In the final section, we apply our results to the substructural *Full Lambek logic* $\mathbf{FL}$ and its extensions [18, 19, 20]. For the commutative case, using the Glivenko nucleus, $\neg\neg$ [3, 6], we recover the aforementioned results in [10, 9, 27]. We also extend our findings to substructural variants of the *open nucleus*, which remain meaningful even in the noncommutative setting.

## 2 Substructural entailment relations

Let $S$ be a set and $S^*$ be the set of finite lists of elements of $S$. Given $\mathbf{L} \subseteq S^* \times S$, we write

$$\mathbf{L} \vdash U \rhd b$$

in place of $(U, b) \in \mathbf{L}$. We often omit "$\mathbf{L} \vdash$" if it is clear from the context.

A *(single-conclusion) substructural entailment relation* is a relation $\mathbf{L} \subseteq S^* \times S$ such that

$$\frac{}{a \rhd a}\,\text{Refl} \qquad\qquad \frac{U \rhd a \qquad V_1, a, V_2 \rhd b}{V_1, U, V_2 \rhd b}\,\text{Trans}$$

These two are the *structural rules* of *reflexivity* and *transitivity*, respectively.

Optional structural rules include *commutativity* (or *exchange*), *monotonicity* (or *weakening*) and *contraction*:

$$\frac{U_1, a_1, a_2, U_2 \rhd b}{U_1, a_2, a_1, U_2 \rhd b}\,\text{Comm} \qquad \frac{U_1, U_2 \rhd b}{U_1, a, U_2 \rhd b}\,\text{Mono} \qquad \frac{U_1, a, a, U_2 \rhd b}{U_1, a, U_2 \rhd b}\,\text{Contr}$$

Quite often an entailment relation $\mathbf{L}$ is inductively generated from a given set $\mathcal{R}$ of rules by closing up with respect to the structural rules above. These *non-structural rules* have the form

$$\frac{U_1 \rhd b_1 \quad \dots \quad U_n \rhd b_n}{U \rhd b}\,r \qquad \text{or, more compactly,} \qquad \frac{\{U_i \rhd b_i \colon 1 \le i \le n\}}{U \rhd b}\,r$$

We call *axioms* the rules that are premissless, i.e. have the form

$$\frac{}{U \rhd b}\, r$$

If $\mathbf{L}$ is inductively generated by a set $\mathcal{R}$ of rules, we write $\mathbf{L} = \langle \mathcal{R} \rangle$.

By an *extension* $\mathbf{L}'$ of an entailment relation $\mathbf{L}$ we mean in general an entailment relation such that $\mathbf{L} \subseteq \mathbf{L}'$. Given an inductively generated entailment relation $\mathbf{L} = \langle \mathcal{R} \rangle$ and a set of rules $\mathcal{R}'$, then we call $\mathbf{L} + \mathcal{R}'$ the entailment relation inductively generated by $\mathcal{R} \cup \mathcal{R}'$. We also say that $\mathbf{L} + \mathcal{R}'$ is an *inductive extension* of $\mathbf{L}$, and we call $\mathcal{R}'$ the set of *additional rules*. Notice that this notation is somewhat informal since $\mathbf{L}$ does not determine $\mathcal{R}$.

## 2.1 Expansive functions and nuclei

▶ **Definition 1.** *Consider the following rules:*

$$\frac{U, a \rhd j(b)}{U, j(a) \rhd j(b)}\, Lj_{\text{left}} \qquad \frac{a, U \rhd j(b)}{j(a), U \rhd j(b)}\, Lj_{\text{right}} \qquad \frac{U, a, V \rhd j(b)}{U, j(a), V \rhd j(b)}\, \mathrm{L}j \qquad \frac{U \rhd b}{U \rhd j(b)}\, Rj$$

*Given a set $S$ endowed with an entailment relation $\mathbf{L}$, we say that a function $j \colon S \to S$ is*

- expansive (over $\mathbf{L}$) *if $Rj$ is derivable in $\mathbf{L}$;*
- *if $j$ is expansive, then it is*
  - *a* right nucleus (over $\mathbf{L}$) *if $Lj_{\text{right}}$ is derivable in $\mathbf{L}$;*
  - *a* left nucleus (over $\mathbf{L}$) *if $Lj_{\text{left}}$ is derivable in $\mathbf{L}$;*
  - *a* bi-nucleus (over $\mathbf{L}$) *if $Lj$ is derivable in $\mathbf{L}$.*

▶ Remark 2. One may wonder why we require the rules to be *derivable* rather than merely *admissible* (see [23, page 21]). Insisting on derivability allows for an immediate and straightforward proof of Proposition 4 below, which states that the property of being a nucleus is inherited by $j$ in the inductive extension. While it is possible to require only admissibility, this would necessitate stricter conditions on the statements that follow and rely on Proposition 4.

For the sake of clarity, we point out that the derivability restriction on the rule $\mathrm{R}j$ is not really restrictive, as it is easily shown that admissibility of $\mathrm{R}j$ suffices to derive the axiom

$$\frac{}{b \rhd j(b)}$$

which, in turn, allows the derivability of $\mathrm{R}j$, since Trans is always taken as a primitive rule. Therefore, the restriction only matters when considering the rules $\mathrm{L}j_{\text{right}}$, $\mathrm{L}j_{\text{left}}$, and $\mathrm{L}j$.

In fact, in most cases, the restriction is not limiting. For instance, if the language includes an implication connective $\to$, then the derivability of the two rules $\mathrm{L}j$ and $\mathrm{R}j$ corresponds to the two directions of the axiom:

$$\frac{}{a \to j(b) \lhd\!\rhd j(a) \to j(b)}$$

and, for axioms, derivability coincides with admissibility [8]. Similar axioms can be formulated in the presence of right and left implication when considering $\mathrm{L}j_{\text{right}}$ and $\mathrm{L}j_{\text{left}}$.

To illustrate that mere admissibility of $\mathrm{L}j$ does not generally suffice for Proposition 4, we provide a toy example. Consider a language with only constants $0, 1$ and a unary operator $\Diamond$. Let

$$\mathbf{L} = \left\langle \left\{ \frac{}{a \rhd \Diamond a}\,\mathrm{Ax1},\ \frac{}{\Diamond\Diamond a \rhd \Diamond a}\,\mathrm{Ax2} \right\} \right\rangle \qquad \text{and} \qquad \mathbf{L}' = \mathbf{L} + \left\{ \frac{}{0 \rhd \Diamond 1}\,\mathrm{Ax3} \right\}.$$

First, note that a straightforward induction proof ensures that all sequents here are of the form $a \rhd b$, where $b$ is either $a$ or something of the form $\Diamond c$.

We claim that $\Diamond$ acts as a bi-nucleus over $\mathbf{L}$ but not over $\mathbf{L}'$.

In both cases, R$j$ is derivable as a consequence of Ax1 (see also Lemma 6.1). We show that

$$\frac{a \rhd \Diamond b}{\Diamond a \rhd \Diamond b}\,\mathrm{L}j$$

is admissible—but not derivable—in $\mathbf{L}$ by induction on the derivation of the premiss. We consider four cases:

1. If the premiss arises from an application of Refl, then it is of the form $\Diamond a \rhd \Diamond a$, and the conclusion is simply an instance of Ax2.
2. If the premiss arises from an application of Ax1, then it is of the form $a \rhd \Diamond a$, and the conclusion follows by an instance of Refl.
3. If the premiss arises from an application of Ax2, then it is of the form $\Diamond\Diamond a \rhd \Diamond a$, and the conclusion results from an application of Trans to the instances $\Diamond\Diamond\Diamond a \rhd \Diamond\Diamond a$ and $\Diamond\Diamond a \rhd \Diamond a$ of Ax2.
4. If the premiss arises from an application of Trans, then we have

$$\frac{a \rhd c \quad c \rhd \Diamond b}{a \rhd \Diamond b}\,\mathrm{Trans}$$

We then have two subcases: when $c = a$, which is trivial, and when $c = \Diamond d$ for some $d$, in which case we apply the induction hypothesis on the left premiss, then apply Trans with the right premiss.

Therefore, $\Diamond$ is a bi-nucleus over $\mathbf{L}$.

If L$j$ were admissible in $\mathbf{L}'$, then we could derive $\mathbf{L}' \vdash \Diamond 0 \rhd \Diamond 1$ by Ax3. However, a simple induction proof ensures that $\Diamond 0 \rhd c$ implies $c = \Diamond^n 0$. Therefore, $\Diamond$ is not a nucleus over $\mathbf{L}'$.

▶ **Lemma 3.**
1. *Every bi-nucleus is both a left and right nucleus.*
2. *If Comm is admissible in* $\mathbf{L}$*, then the definitions of right, left and bi-nucleus coincide.*

**Proof.** Straightforward. ◀

▶ **Proposition 4.** *Let* $\mathbf{L}$ *be an inductively generated entailment relation, and let* $\mathbf{L}'$ *be an inductive extension of* $\mathbf{L}$*. If* $j$ *is an expansive function (respectively (right/left/bi-) nucleus) over* $\mathbf{L}$*, then it also is an expansive function (respectively (right/left/bi-) nucleus) over* $\mathbf{L}'$*.*

**Proof.** If $\mathbf{L}' = \mathbf{L} + \mathcal{R}'$ and there is a derivation $\mathcal{D}$ of a certain rule in $\mathbf{L}$, then $\mathcal{D}$ is a derivation of the same rule also in $\mathbf{L} + \mathcal{R}'$. ◀

▶ **Definition 5.** *Let* $j\colon S \to S$*. Given a rule* $r$*, its* $j$-version $r_j$ *is obtained by putting* $j$ *in front of every succedent:*

$$\frac{\{U_i \rhd b_i \colon i \in I\}}{U \rhd b}\,r \quad \rightsquigarrow \quad \frac{\{U_i \rhd j(b_i) \colon i \in I\}}{U \rhd j(b)}\,r_j$$

*Given a rule* $r$ *that is admissible in* $\mathbf{L}$*, we say that* $r$ survives after $j$ (over $\mathbf{L}$) *if* $r_j$ *is admissible in* $\mathbf{L}$*. We also define the* stability axiom *for* $j$*:*

$$\frac{}{j(a) \rhd a}\,j\text{-}stab$$

▶ **Lemma 6.**

1. $\mathrm{R}j$ *and* $\mathrm{Refl}_j$ *are equivalently admissible and equivalently derivable (in* $\mathbf{L}$*).*
2. $\mathrm{L}j$ *and* $\mathrm{Trans}_j$ *are equivalently admissible and equivalently derivable (in* $\mathbf{L}$*).*
3. $j$-stab *and the following are equivalently admissible and equivalently derivable (in* $\mathbf{L}$*)*

$$\frac{U, a, V \rhd b}{U, j(a), V \rhd b}\,\mathrm{L}j^{+}$$

**Proof.**

1. A conclusion $a \rhd j(a)$ of $\mathrm{Refl}_j$ can be derived by subsequent applications of Refl and $\mathrm{R}j$. On the other hand, if we have the premiss $U \rhd b$ of $\mathrm{R}j$ and by $\mathrm{Refl}_j$ we can derive $b \rhd j(b)$, then by Trans we can derive $U \rhd j(b)$, which is the conclusion of $\mathrm{R}j$.
2. Suppose that we have the premisses $V_1, a, V_2 \rhd j(b)$ and $V_1, j(a), V_2 \rhd j(b)$ of $\mathrm{Trans}_j$. If we apply $\mathrm{L}j$ to the latter, and then apply Trans with the former, we derive $V_1, U, V_2 \rhd j(b)$, which is the conclusion of $\mathrm{Trans}_j$. On the other hand, if we have the premiss $U, a, V \rhd j(b)$ of $\mathrm{L}j$, we can apply $\mathrm{Trans}_j$ to it with $j(a) \rhd j(a)$ to obtain $U, j(a), V \rhd j(b)$, which is the conclusion of $\mathrm{L}j$.
3. Similar to the proof of the first item. ◀

▶ **Definition 7.** *Given an entailment relation* $\mathbf{L} = \langle \mathcal{R} \rangle$ *and an expansive function* $j$ *on* $L$*, we define the following extensions:*

1. *the* Kleisli $j$-extension $\mathrm{K}_j(\mathbf{L})$ *of* $\mathbf{L}$ *by*

$$\mathrm{K}_j(\mathbf{L}) \vdash U \rhd a \iff \mathbf{L} \vdash U \rhd j(a),$$

2. *the* Glivenko $j$-extension $\mathrm{G}_j(\mathbf{L})$ *of* $\mathbf{L}$ *as*

$$\mathrm{G}_j(\mathbf{L}) := \mathbf{L} + \{\mathrm{L}j\} + \{r_j : r \in \mathcal{R}\},$$

3. *the* Kleisli-stable $j$-extension $\mathrm{M}_j(\mathbf{L})$ *of* $\mathbf{L}$ *as*[1]

$$\mathrm{M}_j(\mathbf{L}) := \mathbf{L} + \{j\text{-stab}\}.$$

▶ **Lemma 8.** *Let* $j$ *be an expansive function over* $\mathbf{L}$*. Then:*

1. *All rules derivable in* $\mathbf{L}$ *survive after* $j$ *if and only if* $\mathbf{L} = \mathrm{G}_j(\mathbf{L})$*.*
2. $\mathrm{G}_j(\mathbf{L})$ *and* $\mathrm{M}_j(\mathbf{L})$ *are substructural entailment relations, and* $j$ *is a bi-nucleus over them.*
3. *If* $j$ *is a bi-nucleus over* $\mathbf{L}$*, then* $\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{r_j : r \in \mathcal{R}\}$*.*[2]
4. *If* $j$ *is a bi-nucleus over* $\mathbf{L}$*, then* $\mathrm{K}_j(\mathbf{L})$ *is a substructural entailment relation, and rules* $\mathrm{L}j$ *and* $\mathrm{R}j$ *are admissible.*[3]
5. *If* $\mathbf{L}$ *admits one of the optional structural rules* Comm, Mono, Contr*, then* $\mathrm{K}_j(\mathbf{L})$*,* $\mathrm{G}_j(\mathbf{L})$ *and* $\mathrm{M}_j(\mathbf{L})$ *admit that too.*

**Proof.**

1. Immediate by definition of $\mathrm{G}_j(\mathbf{L})$.

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[1] The notation here is adapted from [9, §8.2]. The "M" stays for "maximum," as $\mathrm{M}_j(\mathbf{L})$ is the greatest extension $\mathbf{L}'$ of $\mathbf{L}$ such that $\mathrm{K}_j(\mathbf{L}') \subseteq \mathrm{M}_j(\mathbf{L})$. See Theorem 11.3, cf. [9, Theorem 8.7].

[2] This is how $\mathrm{G}_j(\mathbf{L})$ is defined in the structural case [8, 7] in which all nuclei are bi-nuclei. However, in the general substructural setting, this property no longer holds and we need to explicitly "force" the rule $\mathrm{L}j$ in the definition of $\mathrm{G}_j(\mathbf{L})$.

[3] This means that "morally" $j$ is a bi-nucleus over $\mathrm{K}_j(\mathbf{L})$. However, since $\mathrm{K}_j(\mathbf{L})$ is not defined inductively, the notion of *derivable rule* (and consequently of *nucleus*) does not make sense in this case.

2. $\mathrm{G}_j(\mathbf{L})$ and $\mathrm{M}_j(\mathbf{L})$ are substructural entailment relations by their very definitions. Moreover, $j$ is expansive also over them as an easy consequence of Proposition 4. We observe that, by definition of $\mathrm{G}_j(\mathbf{L})$, $\mathrm{L}j$ is derivable in it, so $j$ is a bi-nucleus over it. Finally, by Lemma 6.3, we have that $\mathrm{L}j^+$ is derivable in $\mathrm{M}_j(\mathbf{L})$, which let us conclude that $j$ is a bi-nucleus over $\mathrm{M}_j(\mathbf{L})$.
3. Easy consequence of the definition of $\mathrm{G}_j(\mathbf{L})$ and Proposition 4.
4. Refl for $\mathrm{K}_j(\mathbf{L})$ is just $\mathrm{R}j$ for $\mathbf{L}$. If $j$ is a bi-nucleus, then $\mathrm{K}_j(\mathbf{L})$ admits Trans as a direct consequence of Lemma 6.2. While admissibility of $\mathrm{L}j$ is a straightforward consequence of the definition of $\mathrm{K}_j(\mathbf{L})$, we show admissibility of $\mathrm{R}j$:

$$\dfrac{\dfrac{\dfrac{\mathrm{K}_j(\mathbf{L}) \vdash U \rhd b}{\mathbf{L} \vdash U \rhd j(b)}\,\mathrm{defn} \qquad \dfrac{\dfrac{}{\mathbf{L} \vdash j(b) \rhd j(b)}\,\mathrm{Refl}}{\mathbf{L} \vdash j(b) \rhd j(j(b))}\,\mathrm{R}j}{\mathbf{L} \vdash U \rhd j(j(b))}\,\mathrm{Trans}}{\mathrm{K}_j(\mathbf{L}) \vdash U \rhd j(b)}\,\mathrm{defn}$$

5. The claim for $\mathrm{G}_j(\mathbf{L})$ and $\mathrm{M}_j(\mathbf{L})$ readily follows from their very definitions. Also the claim for $\mathrm{K}_j(\mathbf{L})$ is a straightforward consequence of its definition. ◀

▶ Remark 9. Let $j$ be a left/right nucleus over $\mathbf{L}$. Then an easy consequence of $\mathrm{L}j_*$ (where $* \in \{\mathrm{left}, \mathrm{right}\}$), the context-free version of transitivity survives after $j$:

$$\dfrac{a \rhd b \quad b \rhd c}{a \rhd c}\,\mathrm{trans} \qquad \rightsquigarrow \qquad \dfrac{a \rhd j(b) \quad \dfrac{b \rhd j(c)}{j(b) \rhd j(c)}\,\mathrm{L}j_*}{a \rhd j(c)}\,\mathrm{trans}$$

▶ **Proposition 10.** *Let $S$ be a set endowed with an inductively generated entailment relation $\mathbf{L} = \langle \mathcal{R} \rangle$, and let $j$ be expansive over $\mathbf{L}$. Then the following inclusions hold:*

$$\begin{tikzcd} \mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[d, hook] \\ \mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{M}_j(\mathbf{L}) \end{tikzcd}$$

**Proof.**

- $\mathbf{L} \subseteq \mathrm{K}_j(\mathbf{L})$ is nothing but $\mathrm{R}j$.
- $\mathbf{L} \subseteq \mathrm{G}_j(\mathbf{L})$ follows from the definition of $\mathrm{G}_j(\mathbf{L})$.
- $\mathrm{G}_j(\mathbf{L}) \subseteq \mathrm{M}_j(\mathbf{L})$: It suffices to show that the latter admits all rules in the inductive definition of the former. The only non-trivial ones are the $r_j$ where $r \in \mathcal{R}$. Let us consider

$$\dfrac{\{U_i \rhd b_i \colon i \in I\}}{U \rhd b}\,r \qquad \rightsquigarrow \qquad \dfrac{\{U_i \rhd j(b_i) \colon i \in I\}}{U \rhd j(b)}\,r_j$$

We have the following in $\mathrm{M}_j(\mathbf{L})$:

$$\dfrac{\dfrac{\dots \quad \dfrac{U_i \rhd j(b_i) \quad \dfrac{}{j(b_i) \rhd b_i}\,j\text{-stab}}{U_i \rhd b_i}\,\mathrm{Trans} \quad \dots}{U \rhd b}\,r}{U \rhd j(b)}\,\mathrm{R}j$$

- $\mathrm{K}_j(\mathbf{L}) \subseteq \mathrm{M}_j(\mathbf{L})$:

$$\dfrac{\dfrac{\dfrac{\mathrm{K}_j(\mathbf{L}) \vdash U \rhd b}{\mathbf{L} \vdash U \rhd j(b)}\,\text{defn}}{\mathrm{M}_j(\mathbf{L}) \vdash U \rhd j(b)}\,\text{defn} \qquad \dfrac{}{\mathrm{M}_j(\mathbf{L}) \vdash j(b) \rhd b}\,j\text{-stab}}{\mathrm{M}_j(\mathbf{L}) \vdash U \rhd b}\,\text{Trans}$$

◀

## 2.2 The conservation theorem

▶ **Theorem 11** (Conservation). *Let $\mathbf{L} = \langle \mathcal{R} \rangle$, let $\mathbf{L}' = \mathbf{L} + \mathcal{R}'$, and let $j$ be expansive over $\mathbf{L}$. Then:*

$$\mathrm{K}_j(\mathbf{L}') \supseteq \mathbf{L}' \tag{1}$$

$$\mathrm{K}_j(\mathbf{L}') \subseteq \mathbf{L}' \iff \mathbf{L}' \supseteq \mathrm{M}_j(\mathbf{L}) \tag{2}$$

$$\mathrm{K}_j(\mathbf{L}') \subseteq \mathrm{M}_j(\mathbf{L}) \iff \mathbf{L}' \subseteq \mathrm{M}_j(\mathbf{L}) \tag{3}$$

$$\mathrm{K}_j(\mathbf{L}') \supseteq \mathrm{M}_j(\mathbf{L}) \iff \mathbf{L}' \supseteq \mathrm{G}_j(\mathbf{L}) \tag{4}$$

**Proof.**

(1) A particular case of Proposition 10.

(2) If $\mathbf{L}' \supseteq \mathrm{M}_j(\mathbf{L})$, then in particular $\mathbf{L}' \vdash j(a) \rhd a$. Let us suppose $\mathrm{K}_j(\mathbf{L}') \vdash U \rhd a$, i.e. $\mathbf{L}' \vdash U \rhd j(a)$. Then by Trans we conclude $\mathbf{L}' \vdash U \rhd a$. We have shown $\mathrm{K}_j(\mathbf{L}') \subseteq \mathbf{L}'$.
Now suppose that $\mathrm{K}_j(\mathbf{L}') \subseteq \mathbf{L}'$. From $\mathbf{L}' \vdash j(a) \rhd j(a)$, i.e. $\mathrm{K}_j(\mathbf{L}') \vdash j(a) \rhd a$, we get $\mathbf{L}' \vdash j(a) \rhd a$ by assumption. This means that $j$-stab is derivable in $\mathbf{L}'$, which suffices for $\mathbf{L}' \supseteq \mathrm{M}_j(\mathbf{L})$.

(3) The direction "$\Rightarrow$" easily follows from $\mathbf{L}' \subseteq \mathrm{K}_j(\mathbf{L}')$ (Proposition 10).
Now suppose that $\mathbf{L}' \subseteq \mathrm{M}_j(\mathbf{L})$. Since $\mathbf{L} \subseteq \mathbf{L}'$, we have $\mathrm{M}_j(\mathbf{L}) \subseteq \mathrm{M}_j(\mathbf{L}')$. On the other hand, as stability and all the rules of $\mathbf{L}'$ are admissible in $\mathrm{M}_j(\mathbf{L})$, we also have $\mathrm{M}_j(\mathbf{L}') \subseteq \mathrm{M}_j(\mathbf{L})$. Therefore $\mathrm{M}_j(\mathbf{L}) = \mathrm{M}_j(\mathbf{L}')$. From this and $\mathrm{K}_j(\mathbf{L}') \subseteq \mathrm{M}_j(\mathbf{L}')$ (Proposition 10), we conclude $\mathrm{K}_j(\mathbf{L}') \subseteq \mathrm{M}_j(\mathbf{L})$.

(4) Assume $\mathbf{L}' \supseteq \mathrm{G}_j(\mathbf{L})$. Consider one by one the rules that generate $\mathrm{M}_j(\mathbf{L})$: Refl is admissible in $\mathrm{K}_j(\mathbf{L}')$ by an application of R$j$; the optional structural rules are admissible in $\mathrm{K}_j(\mathbf{L}')$ by Lemma 8.5; stability is admissible in $\mathrm{K}_j(\mathbf{L}')$

$$\dfrac{\dfrac{}{\mathbf{L}' \vdash j(a) \rhd j(a)}\,\text{Refl}}{\mathrm{K}_j(\mathbf{L}') \vdash j(a) \rhd a}\,\text{defn}$$

all the rules in $\mathcal{R} \cup \{\text{Trans}\}$, are admissible in $\mathrm{K}_j(\mathbf{L}')$ by Lemma 8.1. As $\mathrm{M}_j(\mathbf{L})$ is the smallest extension of $\mathbf{L}'$ satisfying all these rules, we obtain $\mathrm{K}_j(\mathbf{L}') \supseteq \mathrm{M}_j(\mathbf{L})$.
Assume $\mathrm{K}_j(\mathbf{L}') \supseteq \mathrm{M}_j(\mathbf{L})$. Suppose that $\mathrm{G}_j(\mathbf{L}) \vdash U \rhd a$. We prove $\mathbf{L}' \vdash U \rhd a$ by induction on the last rule used in the derivation of $\mathrm{G}_j(\mathbf{L}) \vdash U \rhd a$.

**a.** Case $r \in \mathcal{R}$: We have

$$\dfrac{\mathrm{G}_j(\mathbf{L}) \vdash U_1 \rhd a_1 \quad \ldots \quad \mathrm{G}_j(\mathbf{L}) \vdash U_n \rhd a_n}{\mathrm{G}_j(\mathbf{L}) \vdash U \rhd a}\,r$$

which leads to

$$\dfrac{\dfrac{\mathrm{G}_j(\mathbf{L}) \vdash U_1 \rhd a_1}{\mathbf{L}' \vdash U_1 \rhd a_1}\,\text{ind. hypothesis} \quad \ldots \quad \dfrac{\mathrm{G}_j(\mathbf{L}) \vdash U_n \rhd a_n}{\mathbf{L}' \vdash U_n \rhd a_n}\,\text{ind. hypothesis}}{\mathbf{L}' \vdash U \rhd a}\,r$$

$$\frac{U, a, V \rhd c \quad U, b, V \rhd c}{U, a \vee b, V \rhd c}\,\mathrm{L}\vee \qquad \frac{U \rhd a}{U \rhd a \vee b}\,\mathrm{R}\vee_1 \qquad \frac{U \rhd b}{U \rhd a \vee b}\,\mathrm{R}\vee_2$$

$$\frac{U, a, V \rhd c}{U, a \wedge b, V \rhd c}\,\mathrm{L}\wedge_1 \qquad \frac{U, b, V \rhd c}{U, a \wedge b, V \rhd c}\,\mathrm{L}\wedge_2 \qquad \frac{U \rhd a \quad U \rhd b}{U \rhd a \wedge b}\,\mathrm{R}\wedge$$

$$\frac{U, V \rhd c}{U, 1, V \rhd c}\,\mathrm{L}1 \qquad \frac{}{\rhd 1}\,\mathrm{R}1$$

$$\frac{U, a, b, V \rhd c}{U, a \cdot b, V \rhd c}\,\mathrm{L}\cdot \qquad \frac{U \rhd a \quad V \rhd b}{U, V \rhd a \cdot b}\,\mathrm{R}\cdot$$

$$\frac{U \rhd a \quad W, b, V \rhd c}{W, U, a \multimap b, V \rhd c}\,\mathrm{L}\multimap \qquad \frac{a, U \rhd b}{U \rhd a \multimap b}\,\mathrm{R}\multimap$$

$$\frac{U \rhd a \quad W, b, V \rhd c}{W, b \multimapinv a, U, V \rhd c}\,\mathrm{L}\multimapinv \qquad \frac{U, a \rhd b}{U \rhd b \multimapinv a}\,\mathrm{R}\multimapinv$$

**Table 1** The rules that generate **FL**, adapted from [9, §2.1].

**b.** Case $r = \mathrm{L}j$ or $r_j$ (where $r \in \mathcal{R}$): We have $a = j(b)$ for some $b \in S$, and hence we can write $\mathrm{G}_j(\mathbf{L}) \vdash U \rhd j(b)$. Since $\mathrm{G}_j(\mathbf{L}) \subseteq \mathrm{M}_j(\mathbf{L}) \subseteq \mathrm{K}_j(\mathbf{L}')$, it follows that $\mathrm{K}_j(\mathbf{L}') \vdash U \rhd j(b)$. By the definition of $\mathrm{K}_j(\mathbf{L}')$, this is equivalent to $\mathbf{L}' \vdash U \rhd j(j(b))$. On the other hand, we observe that $\mathrm{M}_j(\mathbf{L}) \vdash j(j(b)) \rhd b$ by applying $\mathrm{L}j^+$ twice. Then $\mathrm{M}_j(\mathbf{L}) \subseteq \mathrm{K}_j(\mathbf{L}')$ implies $\mathrm{K}_j(\mathbf{L}') \vdash j(j(b)) \rhd b$, which is nothing but $\mathbf{L}' \vdash j(j(b)) \rhd j(b)$. By an application of transitivity between $\mathbf{L}' \vdash U \rhd j(j(b))$ and $\mathbf{L}' \vdash j(j(b)) \rhd j(b)$, we conclude $\mathbf{L}' \vdash U \rhd j(b)$. ◀

We can summarise the main situations in the following:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathbf{L}_1 \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, hook] & \mathbf{L}_2 \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, equal] & \mathbf{L}_3 \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j(\mathbf{L}_1) \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j(\mathbf{L}_2) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j(\mathbf{L}_3)
\end{tikzcd}$$

More detailed diagrams can be found in Appendix A.

## 3 Full Lambek logic and its structural extensions

Fix a set $S$ of formulae, which contains constants $0, 1$ and is closed under connectives $\wedge, \vee, \cdot, \multimapinv, \multimap$. As a basis, we consider *Full Lambek logic* **FL**, which is the substructural entailment relation inductively defined by the rules in Table 1.

To sometimes have a slicker notation, we subsume under $\mathrm{L}\wedge$ and $\mathrm{R}\vee$ the two left rules $\mathrm{L}\wedge_k$ with $k \in \{0, 1\}$ and the two right rules $\mathrm{R}\vee_k$ with $k \in \{0, 1\}$, respectively

Two unary connectives are defined as follows:

$$\sim a := a \multimap 0, \qquad -a := 0 \multimapinv a.$$

We recall a couple of well-known results whose proofs are straightforward:

▶ Remark 12. The following are derivable in **FL**:

$$\frac{}{a, a \multimap b \rhd b}\,\mathrm{mp}{\multimap} \qquad\qquad \frac{}{b \circ\!\!- a, a \rhd b}\,\mathrm{mp}{\circ\!\!-}$$

▶ Remark 13. Let **L** be an inductive extension of **FL**.

1. If $\mathbf{L} \supseteq \{\mathrm{Mono}, \mathrm{Contr}\}$, then $\wedge$ and $\cdot$ coincide.
2. If $\mathbf{L} \supseteq \{\mathrm{Comm}\}$, then $\circ\!\!-$ and $\multimap$ coincide.
3. $\mathrm{R}\wedge$, $\mathrm{R}\circ\!\!-$ and $\mathrm{R}\multimap$ are invertible in **L**.

In $\mathbf{FL}+\{\mathrm{Comm}\}$ and its extensions, we use the usual symbol $\to$ for symmetric implication and the usual symbol $\neg$ for symmetric negation.

An important structural extension is *Minimal logic*, which can be defined as

$$\mathbf{Min} := \mathbf{FL} + \{\mathrm{Comm}, \mathrm{Mono}, \mathrm{Contr}\}.$$

We also define, as usual, *Intuitionistic logic* as $\mathbf{Int} := \mathbf{Min} + \{\mathrm{L0}\}$, where

$$\frac{}{U, 0, V \rhd c}\,\mathrm{L0}$$

▶ Remark 14. In Ono's work, such as [9], the approach differs by allowing the succedent of a sequent to be empty (so that, in particular, in the rules in Fig. 1, $c$ may stand for a formula or nothing.) This framework already includes the following rule within **FL**:

$$\frac{}{0 \rhd}\,\mathrm{L0}$$

Their calculus and our version of **FL** coincide for all sequents where the succedent is non-empty. Specifically, the provability of formulae remains the same, and the underlying logic is unchanged. Through this approach, they obtain **Int** as $\mathbf{FL} + \mathrm{Comm}, \mathrm{Mono}, \mathrm{Mono}_R, \mathrm{Contr}$, where $\mathrm{Mono}_R$ represents right monotonicity:

$$\frac{U \rhd}{U \rhd c}\,\mathrm{Mono}_R$$

## 3.1 Shifts and conservation

Throughout this subsection, let **L** be an inductive extension of **FL**, and let $j$ be a function on formulae. By the *$j$-shift* for the logical operator $*$ we understand the corresponding axiom as in the following table:

| $*$ | **$j$-shift for $*$** |
|---|---|
| $1$ | $1 \rhd j(1)$ |
| $\wedge$ | $j(a) \wedge j(b) \rhd j(a \wedge b)$ |
| $\cdot$ | $j(a) \cdot j(b) \rhd j(a \cdot b)$ |
| $\vee$ | $j(a) \vee j(b) \rhd j(a \vee b)$ |
| $\multimap$ | $a \multimap jb \rhd j(a \multimap b)$ |
| $\circ\!\!-$ | $a \circ\!\!- jb \rhd j(a \circ\!\!- b)$ |

▶ **Theorem 15.** *The right rule* $\mathrm{R}*$ *is admissible for the Kleisli extension* $\mathrm{K}_j(\mathbf{L})$ *if and only if the* $j$*-shift for* $*$ *is derivable for* $\mathbf{L}$*.*

**Proof.** Let us consider the case $* = \wedge$; the other cases are similar.

Suppose first that the $j$-shift for $\wedge$ is derivable in $\mathbf{L}$. If $\mathrm{K}_j(\mathbf{L}) \vdash U \rhd a$ and $\mathrm{K}_j(\mathbf{L}) \vdash U \rhd b$, i.e. $\mathbf{L} \vdash U \rhd j(a)$ and $\mathbf{L} \vdash U \rhd j(b)$, then $\mathrm{R}\wedge$ for $\mathbf{L}$ gives $\mathbf{L} \vdash U \rhd j(a) \wedge j(b)$, which by transitivity and the $j$-shift for $\wedge$ implies $\mathbf{L} \vdash U \rhd j(a \wedge b)$, i.e. $\mathrm{K}_j(\mathbf{L}) \vdash U \rhd a \wedge b$.

As for the converse, assume that $\mathrm{R}\wedge$ is admissible in $\mathrm{K}_j(\mathbf{L})$. By reflexivity, $\mathbf{L} \vdash j(a) \rhd j(a)$ and $\mathbf{L} \vdash j(b) \rhd j(b)$. By $\mathrm{L}\wedge_1$ and $\mathrm{L}\wedge_2$, we get $\mathbf{L} \vdash j(a) \wedge j(b) \rhd j(a)$ and $\mathbf{L} \vdash j(a) \wedge j(b) \rhd j(b)$, i.e. $\mathrm{K}_j(\mathbf{L}) \vdash j(a) \wedge j(b) \rhd a$ and $\mathrm{K}_j(\mathbf{L}) \vdash j(a) \wedge j(b) \rhd b$. Then, $\mathrm{R}\wedge$ for $\mathrm{K}_j(\mathbf{L})$ entails $\mathrm{K}_j(\mathbf{L}) \vdash j(a) \wedge j(b) \rhd a \wedge b$, i.e. $\mathbf{L} \vdash j(a) \wedge j(b) \rhd j(a \wedge b)$. ◀

▶ **Lemma 16.** *Let* $j$ *be a (right/left/bi-) nucleus on* $\mathbf{L}$*. Then*

1. *The* $j$*-shifts for* $1$*,* $\vee$ *are derivable in* $\mathbf{L}$*.*
2. *The* $j$*-shift for* $\cdot$ *is derivable in* $\mathbf{L} + \{\mathrm{Comm}\}$*.*
3. *The* $j$*-shift for* $\wedge$ *is derivable in* $\mathbf{L} + \{\mathrm{Comm}, \mathrm{Mono}, \mathrm{Contr}\}$*.*
4. *If the* $j$*-shift for* $\cdot$ *is derivable, then* $j$ *is a bi-nucleus.*
5. *If* $j$ *is a bi-nucleus, then* $\mathrm{L}\circ\!\!-$ *and* $\mathrm{L}-\!\!\circ$ *survive after* $j$*.*
6. *If the* $j$*-shift for* $\cdot$ *is derivable, then* $\mathrm{L}\circ\!\!-$*,* $\mathrm{L}-\!\!\circ$ *and* Trans *survive after* $j$*.*

**Proof.**

1. The $j$-shift for $1$ is $1 \rhd j(1)$, which is derivable in in $\mathbf{L}$ as a direct consequence of $\mathrm{R}j$.

   We prove that the $j$-shift for $\vee_1$ is derivable in $\mathbf{L}$:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{a \rhd a}\,\mathrm{Refl}}{a \rhd a \vee b}\,\mathrm{R}\vee_1}{a \rhd j(a \vee b)}\,\mathrm{R}j}{j(a) \rhd j(a \vee b)}\,\mathrm{L}j_* \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{b \rhd b}\,\mathrm{Refl}}{b \rhd a \vee b}\,\mathrm{R}\vee_2}{b \rhd j(a \vee b)}\,\mathrm{R}j}{j(b) \rhd j(a \vee b)}\,\mathrm{L}j_*}{j(a) \vee j(b) \rhd j(a \vee b)}\,\mathrm{L}\vee$$

   where $\mathrm{L}j_*$ is either $\mathrm{L}j_{\mathrm{right}}$ or $\mathrm{L}j_{\mathrm{left}}$.

2. We prove that the $j$-shift for $\cdot$ is derivable in $\mathbf{L} + \{\mathrm{Comm}\}$ in the case of $j$ being a right nucleus:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{a \rhd a}\,\mathrm{Refl} \quad \dfrac{}{b \rhd b}\,\mathrm{Refl}}{a, b \rhd a \cdot b}\,\mathrm{R}\cdot}{a, b \rhd j(a \cdot b)}\,\mathrm{R}j}{b, a \rhd j(a \cdot b)}\,\mathrm{Comm}}{j(b), a \rhd j(a \cdot b)}\,\mathrm{L}j_{\mathrm{right}}}{a, j(b) \rhd j(a \cdot b)}\,\mathrm{Comm}}{j(a), j(b) \rhd j(a \cdot b)}\,\mathrm{L}j_{\mathrm{right}}}{j(a) \cdot j(b) \rhd j(a \cdot b)}\,\mathrm{L}\cdot$$

   The case involving a left nucleus is similar.

3. Since $j$-shift for $\cdot$ is derivable in $\mathbf{L}+\{\mathrm{Comm}\}$, it must be derivable in $\mathbf{L}+\{\mathrm{Comm}, \mathrm{Mono}, \mathrm{Contr}\}$ as well (Proposition 4). Then the fact that the $j$-shift for $\wedge$ is derivable in $\mathbf{L} + \{\mathrm{Comm}, \mathrm{Mono}, \mathrm{Contr}\}$ follows from Remark 13.

**4.** Let us prove the case in which $j$ is left, the other is similar. Denote by $\bullet V$ the formula $v_1 \cdot \ldots \cdot v_n$, where $V = v_1, \ldots, v_n$. We show that L$j$ is derivable:

$$\dfrac{\dfrac{\dfrac{}{j(a) \rhd j(a)}\,\mathrm{Refl} \quad \dfrac{\dfrac{}{V \rhd \bullet V}\,(*)}{V \rhd j(\bullet V)}\,\mathrm{R}j}{j(a), V \rhd j(a) \cdot j(\bullet V)}\,\mathrm{R}\wedge \quad \dfrac{\dfrac{}{j(a) \cdot j(\bullet V) \rhd j(a \cdot \bullet V)}\,j\text{-shift}\cdot \quad \dfrac{\dfrac{U, a, V \rhd j(b)}{U, a \cdot \bullet V \rhd j(b)}\,(**)}{U, j(a \cdot \bullet V) \rhd j(b)}\,\mathrm{L}j_{\mathrm{left}}}{U, j(a) \cdot j(\bullet V) \rhd j(b)}\,\mathrm{Trans}}{U, j(a), V \rhd j(b)}\,\mathrm{Trans}$$

where steps $(*)$ and $(**)$ correspond to several subsequent applications of R$\cdot$ and L$\cdot$, respectively, and are straightforwardly provable by induction.

**5.** Let us prove derivability of L$\multimap_j$, the other case is similar.

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{a, a \multimap b \rhd b}\,\mathrm{mp}\multimap}{a, a \multimap b \rhd j(b)}\,\mathrm{R}j}{j(a), a \multimap b \rhd j(b)}\,\mathrm{L}j}{a \multimap b \rhd j(a) \multimap j(b)}\,\mathrm{R}\multimap \quad \dfrac{U \rhd j(a) \quad \dfrac{W, b, V \rhd j(c)}{W, j(b), V \rhd j(c)}\,\mathrm{L}j}{W, U, j(a) \multimap j(b), V \rhd j(c)}\,\mathrm{L}\multimap}{W, U, a \multimap b, V \rhd j(c)}\,\mathrm{Trans}$$

**6.** Follows from the previous two items and Lemma 9. ◀

▶ **Lemma 17.** *Let* $\mathbf{L} = \mathbf{FL} + \mathcal{R}$, *where* $\mathcal{R} \subseteq \{\mathrm{Comm}, \mathrm{Mono}, \mathrm{Contr}\}$, *and let* $j$ *be a (right/left/bi-) nucleus on* $\mathbf{L}$. *Then*

$$\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{j\text{-shift}\cdot, j\text{-shift}\wedge, j\text{-shift}\multimapinv, j\text{-shift}\multimap\}.$$

*If, moreover,* $\mathrm{Comm} \in \mathcal{R}$, *then*

$$\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{j\text{-shift}\wedge, j\text{-shift}\to\}.$$

*If, moreover,* $\mathcal{R} = \{\mathrm{Comm}, \mathrm{Mono}, \mathrm{Contr}\}$, *then*

$$\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{j\text{-shift}\to\}.$$

**Proof.** By definition,

$$\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{\mathrm{L}_j,\ \mathrm{L1}_j,\ \mathrm{R1}_j,\ \mathrm{L}\vee_j,\ \mathrm{R}\vee_{1j},\ \mathrm{R}\vee_{2j},\ \mathrm{L}\wedge_{1j},\ \mathrm{L}\wedge_{2j},\ \mathrm{R}\wedge_j,\ \mathrm{L}\cdot_j,\ \mathrm{R}\cdot_j,$$
$$\mathrm{L}\multimapinv_j,\ \mathrm{R}\multimapinv_j,\ \mathrm{L}\multimap_j,\ \mathrm{R}\multimap_j, [\mathrm{Comm}_j,\ \mathrm{Mono}_j,\ \mathrm{Contr}_j]\},$$

where $\mathrm{Comm}_j$, $\mathrm{Mono}_j$, $\mathrm{Contr}_j$ are present if and only if their counterpart is in $\mathcal{R}$. First, we check that all rules other than R$\wedge$, R$\multimapinv$, R$\multimap$ survive after $j$. We have:

- Rules Comm, Mono, Contr survive after $j$ by Lemma 8.5.
- Rules L1, L$\vee$, L$\wedge_1$, L$\wedge_2$, L$\cdot$ survive after $j$ since they do not change succedents.
- Rules R1, R$\vee_1$, R$\vee_2$ survive after $j$ by Theorem 15 since their $j$-shifts are derivable by Lemma 16.

Therefore we have $\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{\mathrm{R}\cdot_j, \mathrm{R}\wedge_j, \mathrm{R}\multimapinv_j, \mathrm{R}\multimap_j, \mathrm{L}\multimap_j, \mathrm{L}\multimapinv_j\}$. Observe that admissibilities of R$\wedge_j$, R$\cdot_j$, R$\multimapinv_j$ and R$\multimap_j$ are equivalent to admissibilities of $j$-shift$\wedge$, $j$-shift$\cdot$, $j$-shift$\multimapinv$, $j$-shift$\multimap$, respectively, by Theorem 15. By Lemma 16(6), we know that $j$-shift$\cdot$ suffices also for L$\multimap_j$, L$\multimapinv_j$ and Trans$_j$, which is tantamount to L$j$ by Lemma 6(2). Therefore we have $\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{j\text{-shift}\cdot, j\text{-shift}\wedge, j\text{-shift}\multimapinv, j\text{-shift}\multimap\}$.

If, moreover, Comm $\in \mathcal{R}$, then $j$-shift$\cdot$ is derivable by Lemma 16; and $\multimap$ and $\circ\!\!-$ coincide by Remark 13. Therefore we have $\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{j\text{-shift}\wedge, j\text{-shift}\to\}$.

If, moreover, $\mathcal{R} = \{\text{Comm, Mono, Contr}\}$, then $j$-shift$\wedge$ is derivable by Lemma 16. Therefore we conclude $\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{j\text{-shift}\to\}$. ◀

▶ Remark 18. A common definition of a nucleus, as given by Johnstone [16, §C1.1], consists of the following conditions:

1. $a \rhd j(a)$,
2. $j(a) \lhd\!\rhd j(j(a))$,
3. $j(a \cdot b) \lhd\!\rhd j(a) \cdot j(b)$.

Conditions 1 and 2 follow directly from the rules L$j$ and R$j$. Moreover, as a consequence of Lemma 17, we can observe that if all structural rules are satisfied, then condition 3 also holds. However, if some structural rules are omitted, condition 3 may not hold. Thus, in the substructural setting, these two notions of nucleus may differ.

▶ **Theorem 19.** *Let $j$ be a (right/left/bi-) nucleus over* **FL**, *and let* **L** *be an inductive extension of* **FL** *such that* $\mathbf{L} \subseteq \mathrm{M}_j(\mathbf{FL})$. *Then the following are equivalent:*

1. $\mathrm{M}_j(\mathbf{FL}) \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash U \rhd ja$.
2. $\mathrm{G}_j(\mathbf{FL}) \subseteq \mathbf{L}$.
3. *The following conditions hold:*
   - $\mathbf{L} \vdash ja \cdot jb \rhd j(a \cdot b)$,
   - $\mathbf{L} \vdash ja \wedge jb \rhd j(a \wedge b)$,
   - $\mathbf{L} \vdash a \multimap jb \rhd j(a \multimap b)$,
   - $\mathbf{L} \vdash jb \circ\!\!- a \rhd j(b \circ\!\!- a)$.

**Proof.** The equivalence between the first two items is an application of Theorem 11(4). The equivalence between the last two items is an application of Lemma 17. ◀

## 3.2 The Glivenko nuclei

In this section we present a version of the *Glivenko nucleus* [3, 6], i.e. $a \mapsto \neg\neg a$, that works in substructural logic.

▶ **Proposition 20.**

1. *The* left Glivenko nucleus $a \mapsto \sim\!-a$ *is a left nucleus over* **FL***;*
2. *The* right Glivenko nucleus $a \mapsto -\!\sim a$ *is a right nucleus over* **FL**.

**Proof.** We only give the proof for $\sim\!-$, the proof for $-\!\sim$ is similar. Derivability of L$\sim\!-_{\text{left}}$ R$\sim\!-$:

$$\dfrac{\dfrac{\dfrac{}{0 \rhd 0}\,\text{Refl} \qquad \dfrac{\dfrac{U, a \rhd \sim\!-b \qquad \dfrac{\dfrac{}{0 \rhd 0}\,\text{Refl} \quad \dfrac{}{-b \rhd -b}\,\text{Refl}}{-b, \sim\!-b \rhd 0}\,\text{L}\multimap}{-b, U, a \rhd 0}\,\text{Trans}}{-b, U \rhd -a}\,\text{R}\circ\!\!-}{-b, U, \sim\!-a \rhd 0}\,\text{L}\multimap}{U, \sim\!-a \rhd \sim\!-b}\,\text{R}\multimap \qquad \dfrac{\dfrac{\dfrac{}{0 \rhd 0}\,\text{Refl} \quad U \rhd a}{-a, U \rhd 0}\,\text{L}\circ\!\!-}{U \rhd \sim\!-a}\,\text{R}\multimap$$

◀

▶ **Theorem 21.**

1. *Let* **L** *be an inductive extension of* **FL** *such that* $\mathbf{L} \subseteq \mathrm{M}_{\sim\!-}(\mathbf{FL})$. *Then the following are equivalent:*

**a.** $\mathrm{M}_{\sim-}(\mathbf{FL}) \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash U \rhd \sim\!-a$.

**b.** $\mathrm{G}_{\sim-}(\mathbf{FL}) \subseteq \mathbf{L}$.

**c.** *The following conditions hold:*

- $\mathbf{L} \vdash (\sim\!-a) \wedge (\sim\!-b) \rhd \sim\!-(a \wedge b)$,
- $\mathbf{L} \vdash a \multimap \sim\!-b \rhd \sim\!-(a \multimap b)$,
- $\mathbf{L} \vdash \sim\!-b \multimapinv a \rhd \sim\!-(b \multimapinv a)$,

**2.** *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{FL}$ *such that* $\mathbf{L} \subseteq \mathrm{M}_{-\sim}(\mathbf{FL})$. *Then the following are equivalent:*

**a.** $\mathrm{M}_{-\sim}(\mathbf{FL}) \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash U \rhd -\!\sim a$.

**b.** $\mathrm{G}_{-\sim}(\mathbf{FL}) \subseteq \mathbf{L}$.

**c.** *The following conditions hold:*

- $\mathbf{L} \vdash (-\!\sim a) \wedge (-\!\sim b) \rhd -\!\sim(a \wedge b)$,
- $\mathbf{L} \vdash a \multimap -\!\sim b \rhd -\!\sim(a \multimap b)$,
- $\mathbf{L} \vdash -\!\sim b \multimapinv a \rhd -\!\sim(b \multimapinv a)$,

**Proof.** Follows from Theorem 19 and the fact that $\sim\!-$-shift$\cdot$ and $-\!\sim$-shift$\cdot$ are derivable in $\mathbf{FL}$. Let us give the proof of the former, the latter is similar:

$$
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{0 \rhd 0}\,\mathrm{Refl} \quad \dfrac{\dfrac{\dfrac{\dfrac{}{0 \rhd 0}\,\mathrm{Refl} \quad \dfrac{\dfrac{\dfrac{}{0 \rhd 0}\,\mathrm{Refl} \quad \dfrac{\dfrac{\dfrac{}{a \rhd a}\,\mathrm{Refl} \quad \dfrac{}{b \rhd b}\,\mathrm{Refl}}{a, b \rhd a \cdot b}\,\mathrm{R}\cdot}{-(a \cdot b), a, b \rhd 0}\,\mathrm{L}\multimapinv}{-(a \cdot b), a \rhd -b}\,\mathrm{R}\multimapinv}{-(a \cdot b), \sim\!-b, a \rhd 0}\,\mathrm{L}\multimap}{-(a \cdot b), \sim\!-b \rhd -a}\,\mathrm{R}\multimapinv}{-(a \cdot b), \sim\!-a, \sim\!-b \rhd 0}\,\mathrm{L}\multimap}{\sim\!-a, \sim\!-b \rhd \sim\!-(a \cdot b)}\,\mathrm{R}\multimap}{\sim\!-a \cdot \sim\!-b \rhd \sim\!-(a \cdot b)}\,\mathrm{L}\cdot
$$

◀

If we consider the *Full Lambek logic with exchange* $\mathbf{FL}_e := \mathbf{FL} + \{\mathrm{Comm}\}$ [9, §2.1], then the *Glivenko nucleus* $\neg\neg$ is a nucleus over $\mathbf{FL}_e$.

We notice that the extension

$$\mathrm{M}_{\neg\neg}(\mathbf{FL}_e) = \mathbf{FL}_e + \{\neg\neg\text{-stab}\},$$

corresponds to the one known in literature as the *Involutive Full Lambek logic with exchange* $\mathbf{InFL}_e$ [9, §7.3]. If, moreover, we consider $\mathbf{Min}$, we obtain two well-known extensions: $\mathbf{Cla} := \mathrm{M}_{\neg\neg}(\mathbf{Min})$ is *Classical logic* and $\mathbf{Gli} := \mathrm{G}_{\neg\neg}(\mathbf{Min})$ is the *Glivenko logic* [9, 26, 32].

▶ **Corollary 22.**

**1.** (Ono [27]) *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{FL}_e$ *such that* $\mathbf{L} \subseteq \mathbf{InFL}_e$. *Then the following are equivalent:*

**a.** $\mathbf{InFL}_e \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash U \rhd \neg\neg a$.

**b.** $\mathrm{G}_{\neg\neg}(\mathbf{FL}_e) \subseteq \mathbf{L}$.

**c.** *The following conditions hold:*

- $\mathbf{L} \vdash \neg\neg a \wedge \neg\neg b \rhd \neg\neg(a \wedge b)$,
- $\mathbf{L} \vdash a \to \neg\neg b \rhd \neg\neg(a \to b)$.

2. (Odintsov [24, 25])[4] *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{Min}$ *such that* $\mathbf{L} \subseteq \mathbf{Cla}$. *Then the following are equivalent:*
   a. $\mathbf{Cla} \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash U \rhd \neg\neg a$.
   b. $\mathbf{Gli} \subseteq \mathbf{L}$.
   c. $\mathbf{L} \vdash a \to \neg\neg b \rhd \neg\neg(a \to b)$.
3. (Glivenko [11, 12]) $\mathbf{Cla} \vdash U \rhd a$ *if and only if* $\mathbf{Int} \vdash U \rhd \neg\neg a$.

**Proof.** Particular cases of 21. The only thing left to show is $\mathbf{Int} \vdash a \to \neg\neg b \rhd \neg\neg(a \to b)$, which is well-known (see, e.g. [8]). ◀

## 3.3 The open nuclei

In this section we present a version of the *open nucleus* [3, 6], i.e. $a \mapsto (0 \to a)$, that works in substructural logic.

▶ **Proposition 23.**
1. *The* right open nucleus $o_R : a \mapsto 0 \multimap 0 \cdot a$ *is a right nucleus over* $\mathbf{FL}$*;*
2. *The* left open nucleus $o_L : a \mapsto a \cdot 0 \mathrel{\circ\!\!-} 0$ *is a left nucleus over* $\mathbf{FL}$.

**Proof.** We only give the prove for $o_R$, the proof for $o_L$ is similar. Derivability of $\mathrm{R}o_R$:

$$\dfrac{\dfrac{\dfrac{}{0 \rhd 0}\,\mathrm{Refl} \qquad U \rhd a}{0, U \rhd 0 \cdot a}\,\mathrm{R}\cdot}{U \rhd 0 \multimap 0 \cdot a}\,\mathrm{R}\multimap$$

Before proving derivability of $\mathrm{L}o_{R\mathrm{right}}$, we need:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{0 \rhd 0}\,\mathrm{Refl} \qquad \dfrac{\dfrac{\dfrac{}{a \rhd a}\,\mathrm{Refl} \qquad \dfrac{\dfrac{}{0 \rhd 0}\,\mathrm{Refl} \qquad \dfrac{}{0 \cdot b \rhd 0 \cdot b}\,\mathrm{Refl}}{0, 0 \multimap 0 \cdot b \rhd 0 \cdot b}\,\mathrm{L}\multimap}{0, a, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \cdot b}\,\mathrm{L}\multimap}{0 \cdot a, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \cdot b}\,\mathrm{L}\cdot}{0, 0 \multimap 0 \cdot a, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \cdot b}\,\mathrm{L}\multimap}{0 \multimap 0 \cdot a, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \multimap 0 \cdot b}\,\mathrm{R}\multimap}{a \multimap (0 \multimap 0 \cdot b) \rhd (0 \multimap 0 \cdot a) \multimap (0 \multimap 0 \cdot b)}\,\mathrm{R}\multimap \qquad \dfrac{\dfrac{}{0 \multimap 0 \cdot a \rhd 0 \multimap 0 \cdot a}\,\mathrm{Refl} \qquad \dfrac{}{0 \multimap 0 \cdot b \rhd 0 \multimap 0 \cdot b}\,\mathrm{Refl}}{0 \multimap 0 \cdot a, (0 \multimap 0 \cdot a) \multimap (0 \multimap 0 \cdot b) \rhd 0 \multimap 0 \cdot b}\,\mathrm{L}\multimap}{0 \multimap 0 \cdot a, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \multimap 0 \cdot b}\,\mathrm{Trans}$$

Derivability of $\mathrm{L}o_{R\mathrm{right}}$:

$$\dfrac{\dfrac{a, U \rhd 0 \multimap 0 \cdot b}{U \rhd a \multimap (0 \multimap 0 \cdot b)}\,\mathrm{R}\multimap \qquad \dfrac{}{0 \multimap 0 \cdot a, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \multimap 0 \cdot b}}{0 \multimap 0 \cdot a, U \rhd 0 \multimap 0 \cdot b}\,\mathrm{Trans}$$

◀

▶ **Theorem 24.**
1. *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{FL}$ *such that* $\mathbf{L} \subseteq \mathrm{M}_{o_R}(\mathbf{FL})$. *Then the following are equivalent:*
   a. $\mathrm{M}_{o_R}(\mathbf{FL}) \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash 0, U \rhd 0 \cdot a$.
   b. $\mathrm{G}_{o_R}(\mathbf{FL}) \subseteq \mathbf{L}$.
   c. *The following conditions hold:*
      - $\mathbf{L} \vdash 0, (0 \multimap 0 \cdot a) \cdot (0 \multimap 0 \cdot b) \rhd 0 \cdot (a \cdot b)$,

[4] This was proved by Odintsov [24] and shortly after by Cignoli and Torres [5]. This fact was already noted by Segerberg, who mentioned it on [32, page 46] without providing any proof or further reference. In fact, the first published proofs that we are aware of are the two mentioned above.

- $\mathbf{L} \vdash 0, (0 \multimap 0 \cdot a) \wedge (0 \multimap 0 \cdot b) \rhd 0 \cdot (a \wedge b)$,
- $\mathbf{L} \vdash 0, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \cdot (a \multimap b)$,
- $\mathbf{L} \vdash 0, (0 \multimap 0 \cdot b) \circ\!\!- a \rhd 0 \cdot (b \circ\!\!- a)$,

**2.** *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{FL}$ *such that* $\mathbf{L} \subseteq \mathrm{M}_{o_L}(\mathbf{FL})$. *Then the following are equivalent:*

**a.** $\mathrm{M}_{o_L}(\mathbf{FL}) \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash U, 0 \rhd a \cdot 0$.

**b.** $\mathrm{G}_{o_L}(\mathbf{FL}) \subseteq \mathbf{L}$.

**c.** *The following conditions hold:*

- $\mathbf{L} \vdash (a \cdot 0 \circ\!\!- 0) \cdot (b \cdot 0 \circ\!\!- 0), 0 \rhd (a \cdot b) \cdot 0$,
- $\mathbf{L} \vdash (a \cdot 0 \circ\!\!- 0) \wedge (b \cdot 0 \circ\!\!- 0), 0 \rhd (a \wedge b) \cdot 0$,
- $\mathbf{L} \vdash a \multimap (b \cdot 0 \circ\!\!- 0), 0 \rhd (a \multimap b) \cdot 0$,
- $\mathbf{L} \vdash (b \cdot 0 \circ\!\!- 0) \circ\!\!- a, 0 \rhd (b \circ\!\!- a) \cdot 0$,

**Proof.** Follows from Theorem 19 and invertibility of rules R$\multimap$ and R$\circ\!\!-$ (Remark 13). ◀

## 4 Future work

In this paper, we have focused on the provability-theoretic aspects of Glivenko-type theorems and $j$-shift admissibility within substructural entailment relations. However, our approach has remained *proof-irrelevant* in that it treats derivability relations extensionally, without distinguishing between different proofs of the same judgment.

A natural direction for future work is to explore the same framework from a *proof-relevant* perspective. This would involve enriching the underlying entailment relations to retain information about the structure or identity of proofs, such as labelling derivations, formalising proof terms, or working within a categorical semantics where morphisms represent proofs explicitly.

Several questions arise in this setting:

- How do Glivenko extensions interact with proof equivalences? Given an equivalence relation on proofs in an entailment relation $\mathbf{L}$, can one define a canonical transformation between proofs of $\mathbf{L} \vdash U \rhd a$ and proofs of $\mathrm{G}_j(\mathbf{L}) \vdash U \rhd j(a)$ in a way that preserves the equivalence relation? Moreover, if such a transformation exists, does it have an inverse? This question is related to the decorations studied in Schellinx's PhD thesis [31], where the codomain of the translation is classical linear logic, while here we are in a more abstract setting.
- In what sense are $j$-shifts invertible or conservative at the level of proofs, not just derivability? What kinds of monads arise categorically?
- Can we extend our results to a setting like type theory or logical frameworks (e.g., LF [13], FOLDS [21]), where proofs themselves are first-class objects?

Adopting a proof-relevant viewpoint could also facilitate connections with categorical logic and type theory. Additionally, this perspective might yield new insights into normalisation, proof transformation, or computational interpretations of substructural logics.

We leave these directions for future investigation.

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# Appendix

## A Diagrams of extensions

Supposing that $L \subsetneq \mathrm{G}_j(\mathbf{L}) \subsetneq \mathrm{M}_j(\mathbf{L})$:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L})
\end{tikzcd}$$

We want to see what happens if we add some inductive extension $\mathbf{L}'$ of $\mathbf{L}$ to the diagram.

1. Case $\mathbf{L}'$ unrelated to $\mathrm{M}_j(\mathbf{L})$:

$$\begin{tikzcd}
\mathbf{L} \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}) \arrow[dr, hook] \arrow[dd, equal] & \\
& \mathbf{L}' \arrow[rr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}') \arrow[rr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[rr, equal] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, hook] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[rr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

2. Case $\mathbf{L}' \subsetneq \mathrm{M}_j(\mathbf{L})$ but unrelated to $\mathrm{G}_j(\mathbf{L})$:

$$\begin{tikzcd}
\mathbf{L} \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] \arrow[dd, equal] & \\
& \mathbf{L}' \arrow[rr, hook] \arrow[dd, hook] \arrow[urrr, hook] & & \mathrm{G}_j(\mathbf{L}') \arrow[rr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[rr, equal] \arrow[dr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[rr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

3. Case $\mathbf{L} \subsetneq \mathbf{L}' \subsetneq \mathrm{G}_j(\mathbf{L})$:

$$\begin{tikzcd}
\mathbf{L} \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, equal] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] \arrow[dd, equal] & \\
& \mathbf{L}' \arrow[rr, hook] \arrow[dd, hook] \arrow[urr, hook] & & \mathrm{G}_j(\mathbf{L}') \arrow[rr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[rr, equal] \arrow[dr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[rr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

Simplified:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathbf{L}' \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}') \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, equal] & \mathrm{M}_j(\mathbf{L}') \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j(\mathbf{L}') \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

**4.** Case $\mathrm{G}_j(\mathbf{L}) \subsetneq \mathbf{L}' \subsetneq \mathrm{M}_j(\mathbf{L})$:

$$\begin{tikzcd}
\mathbf{L} \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] \arrow[dd, equal] \arrow[ddll, hook] & \\
& \mathbf{L}' \arrow[rr, equal] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}') \arrow[rr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[rr, equal] \arrow[dr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

Simplified:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, hook] & \mathbf{L}' \arrow[r, equal] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}') \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, equal] & \mathrm{M}_j(\mathbf{L}') \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

**5.** Case $\mathrm{M}_j(\mathbf{L}) \subsetneq \mathbf{L}'$:

$$\begin{tikzcd}
\mathbf{L} \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}) \arrow[dr, hook] \arrow[dd, equal] \arrow[ddll, hook] & \\
& \mathbf{L}' \arrow[rr, equal] \arrow[dd, equal] & & \mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] \arrow[dd, equal] & & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[rr, equal] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, hook] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

Simplified:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, equal] & \mathbf{L}' \arrow[r, equal] \arrow[d, equal] & \mathrm{G}_j(\mathbf{L}') \arrow[r, equal] \arrow[d, equal] & \mathrm{M}_j(\mathbf{L}') \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

**6.** Case $\mathbf{L}' = \mathbf{L}$: Trivial.

**7.** Case $\mathbf{L}' = \mathrm{G}_j(\mathbf{L})$:

$$
\begin{tikzcd}
\mathbf{L} \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, equal] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] \arrow[dd, equal] & \\
& \mathbf{L}' \arrow[rr, equal] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}') \arrow[rr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[rr, equal] \arrow[dr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}
$$

Simplified:

$$
\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, hook] & \mathrm{G}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, equal] & \mathrm{M}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{G}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j\,\mathrm{G}_j(\mathbf{L})
\end{tikzcd}
$$

**8.** Case $\mathbf{L}' = \mathrm{M}_j(\mathbf{L})$:

$$
\begin{tikzcd}
\mathbf{L} \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] \arrow[dd, equal] & \\
& \mathbf{L}' \arrow[rr, equal] \arrow[dd, equal] & & \mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] \arrow[dd, equal] & & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[rr, equal] \arrow[dr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}
$$

Simplified:

$$
\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, equal] & \mathrm{G}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, equal] & \mathrm{M}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{G}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j\,\mathrm{M}_j(\mathbf{L})
\end{tikzcd}
$$